

1.1) Finite Difference Approximation of E

forward differences: $\|\nabla u_{i,j}\|_2^2 = (u_{i+1,j} - u_{i,j})^2 + (u_{i,j+1} - u_{i,j})^2$

$$E[u] = \sum_{i,j} \Omega[i,j] (u_{i,j} - g_{i,j})^2 + \lambda \sum_{i,j} [(u_{i+1,j} - u_{i,j})^2 + (u_{i,j+1} - u_{i,j})^2]$$

1.2) Gradient of Discrete Energy

$$u_c[i,j] = u_{i,j}$$

$$\frac{\partial}{\partial u_{i,j}} \left(\sum_{i,j} \Omega[i,j] (u_{i,j} - g_{i,j})^2 \right) = 2 \Omega[i,j] (u_{i,j} - g_{i,j})$$

$$(i,j) ; i=0, \dots, M ; j=0, \dots, N$$

$$\frac{\partial}{\partial u_{i,j}} (u_{i+1,j} - u_{i,j})^2 = (2u_{i,j} - u_{i+1,j} - u_{i-1,j})$$

$$\frac{\partial}{\partial u_{i,j}} (u_{i,j+1} - u_{i,j})^2 = (2u_{i,j} - u_{i,j+1} - u_{i,j-1})$$

$$\nabla_u E(u) = 2 \Omega[i,j] (u_{i,j} - g_{i,j})^2 + 2\lambda (4u_{i,j} - u_{i+1,j} - u_{i-1,j} - u_{i,j+1} - u_{i,j-1}) \quad \forall i \in [2, M-1], \forall j \in [2, N-1]$$

Bottom

Top edge: $2 \Omega_{1,j} (u_{1,j} - g_{1,j}) + 2\lambda (3u_{1,j} - u_{2,j} - u_{1,j+1} - u_{1,j-1})$ $\begin{matrix} i=1 \\ j \in [2, N-1] \end{matrix}$

Bottom edge: $2 \Omega_{M,j} (u_{M,j} - g_{M,j}) + 2\lambda (3u_{M,j} - u_{M-1,j} - u_{M,j+1} - u_{M,j-1})$ $\begin{matrix} i=M \\ j \in [2, N-1] \end{matrix}$

Left edge: $2 \Omega_{i,1} (u_{i,1} - g_{i,1}) + 2\lambda (3u_{i,1} - u_{i,2} - u_{i+1,1} - u_{i-1,1})$ $\begin{matrix} i \in [2, M-1] \\ j=1 \end{matrix}$

Right edge: $2 \Omega_{i,N} (u_{i,N} - g_{i,N}) + 2\lambda (3u_{i,N} - u_{i,N-1} - u_{i+1,N} - u_{i-1,N})$ $\begin{matrix} i \in [2, M-1] \\ j=N \end{matrix}$

Top left: $2 \Omega_{1,1} (u_{1,1} - g_{1,1}) + 2\lambda (2u_{1,1} - u_{2,1} - u_{1,2})$

Top right: $2 \Omega_{1,N} (u_{1,N} - g_{1,N}) + 2\lambda (2u_{1,N} - u_{2,N} - u_{1,N-1})$

Bottom left: $2 \Omega_{M,1} (u_{M,1} - g_{M,1}) + 2\lambda (2u_{M,1} - u_{M,2} - u_{M-1,1})$

Bottom right: $2 \Omega_{M,N} (u_{M,N} - g_{M,N}) + 2\lambda (2u_{M,N} - u_{M-1,N} - u_{M,N-1})$